1000000 raw units can mean 1 token

Frontend dApp Engineer · 6-decimal example · Five answers

FE–F–02 | Foundation | Practical

How do you format a number from a `uint256` value, like an ERC-20 token balance, for display in a UI?

Answer summary · authored

For a token with 6 decimals, viem formatUnits(1000000n, 6) returns '1'.

Reading chain state and authorizing actions

FE-F-01 | Foundation | Knowledge

What is a"provider" in Ethers.js or Viem and what is its role?

Answer summary · authored

In ethers v6, a Provider reads chain data; a Signer authorizes transactions. In viem, Public Client exposes reads and Wallet Client handles signing via a wallet or local account.

Format units without losing integer precision

FE-F-02 | Foundation | Practical

How do you format a number from a `uint256` value, like an ERC-20 token balance, for display in a UI?

Answer summary · authored

Use formatUnits(balance, decimals), keeping balance as bigint. In viem, formatUnits(1000000n, 6) returns '1'. Get decimals for that token and chain; never assume 18.

ERC-20 decimals() is optional. If absent, use verified token metadata or label raw units instead of silently assuming 18.

A cancellation races the pending transaction

FE-I-01 | Intermediate | Practical

A user reports their transaction is"stuck" as pending. What are the possible causes and how could you handle this in the UI?

Answer summary · authored

Low fees or an earlier pending nonce can block it. Show pending status. A speed-up uses the same nonce with higher fees; cancellation is a competing replacement, not a guarantee.

NFT ownership needs an index

FE-A-01 | Advanced | Architecture

How would you design a frontend to display a user's entire NFT collection, including images and metadata, across multiple contracts?

Answer summary · authored

Query an ownership indexer by wallet and chain, paginate results, then render token IDs, metadata and images. Handle missing media. A balanceOf call cannot list every collection.

Typed signatures still need replay controls

FE-E-01 Expert Design

You are building a dApp that requires users to sign messages for off-chain actions (e.g., voting, placing orders on an off-chain order book). Describe the EIP-712 standard and explain why it is superior to signing a simple string with `personal_sign`.

Answer summary · authored

It signs typed fields with a domain, so wallets can show structured terms. Include chainId and verifyingContract. The app must still enforce nonces and expiry.